

Technical Specifications: Solid-State Battery Architecture

1. Anode Structure: Fractal Graphene Mesh & Passive Dendrite Management

This section details the maximum-performance configuration for the anode architecture, engineered to completely neutralize dendrite formation even under extreme industrial loads and hyper-fast charging cycles.

1.1 Fractal Honeycomb Architecture

- **Material:** High-Density Graphene Oxide (GO) synthesized via high-yield Plasma CVD.
- **Geometry:** Non-linear, mathematically optimized fractal labyrinth structure.
- **Mechanism (Passive Dendrite Management):** Instead of utilizing a rigid barrier, this structure guides nascent lithium dendrite growth into electrically neutral channels. The dendrites are trapped and forced into electrochemical dissolution before they can bridge the gap to the cathode.

1.2 Self-Healing Interface (Molecular Bridge)

- **Material:** Gallium-Indium (Ga-In) liquid metal alloy.
- **Function:** Acts as an atomic-level shock absorber at the interface. Localized mechanical stress from ion deposition temporarily alters the localized viscosity, absorbing pressure and preventing structural fracturing of the solid state.

2. Electrolyte Core: The Bio-Mimetic Gradient

To achieve peak ionic conductivity without sacrificing mechanical integrity, the system utilizes a proprietary dual-phase gradient matrix, operating at the theoretical maximum of ion transport efficiency.

2.1 Sintered LLZO Framework

- **Material:** Lithium Lanthanum Zirconate (LLZO) ceramic fibers.
- **Performance:** Sintered into a directional mesh, these fibers provide ultra-high-speed ion transport channels, maximizing charge and discharge rates while maintaining structural rigidity.

2.2 Succinonitrile (SN) Matrix Infiltration

- **Material:** Organic plastic crystals (Succinonitrile).
- **Integration:** These crystals are infiltrated into the microscopic voids of the ceramic structure. They behave structurally as a solid but offer liquid-like molecular mobility.
- **Result:** Interfacial resistance is reduced to near absolute zero, allowing seamless electron and ion flow.

3. Thermal Management: Directed Heat Dissipation

Designed for absolute thermal stability under maximum operative stress, ensuring inherent safety and non-flammability.

3.1 Hexagonal Boron Nitride (h-BN) Nanoplatelets

- **Material:** Cost-effective, high-yield "White Graphene" (h-BN).
- **Properties:** Delivers exceptional thermal conductivity (up to 600 W/mK) while acting as a perfect electrical insulator, entirely eliminating the risk of internal short circuits while dissipating massive heat loads.

3.2 Electrostatic Additive Manufacturing

- **Manufacturing Process:** 3D Structural Printing with dynamic gradient adjustment.
- **Orientation:** During extrusion, an active electrostatic field aligns the h-BN nanoplatelets perfectly perpendicular to the core heat generation zones.
- **Result:** The creation of highly efficient, vertical "heat highways" that instantaneously transport thermal spikes directly to the outer thermal casing, eliminating any possibility of thermal runaway.